

# Identifying Personal Network Types: Structural Topologies, Empirical Decision Rules, and Longitudinal Transitions in a Collegiate Setting

David Hachen\* Omar Lizardo† Brandon Sepulvado‡ Cheng Wang§

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## Abstract

Building on foundational work by [Bidart et al. \(2018\)](#) and [Vacca \(2020\)](#), we investigate whether ego networks can be classified into discrete types based on their structural signatures using eight survey waves from the NetHealth Study ( $N = 701$ ). Using unsupervised  $k$ -means clustering on ego-network topological metrics, we recover four configurations corresponding to types discussed in previous work: “Pearl Collar Networks” (large, modular, high-diameter), “Segmented Networks” (moderately sized, decentralized), “Centered Star Networks” (highly centralized around key intermediaries), and “Regular Dense Networks” (small, cohesive, tightly knit). With this classification in hand, we extend prior scholarship in three major directions. First, we use empirical classification trees to predict individual membership in each ego-network type with 92.9% accuracy, establishing explicit, data-driven cutoffs for values of each structural signature to determine membership in each type. Second, we use longitudinal Markov models across 1,900 transitions to examine the dynamic evolution of individuals across types, revealing high persistence in regular dense network structures (70.8%) alongside substantial structural mobility across the other types. Third, we evaluate alter-level functional support provision across four domains (companionship, advice, emotional comfort, and financial assistance) alongside tie multiplexity and interpersonal trust ( $N = 580$ ),

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\*Department of Sociology, University of Notre Dame.

†Department of Sociology, University of California, Los Angeles.

‡NORC.

§Department of Sociology, Wayne State University.

showing how topological structure systematically governs relational bandwidth and emotional solidarity.

**Keywords:** Personal networks, egocentric network analysis, structural typologies, classification trees, Markov models, social support, relational multiplexity.

# 1 Introduction

The study of ego networks occupies a central position in network analysis, linking macro-level institutional configurations to micro-level interpersonal interactions (Smith, 2020; Perry et al., 2018). Early efforts to categorize personal networks were predominantly attribute-based, classifying networks according to role relationships, such as the balance between kin and non-kin, or the specific forms of social support exchanged across ties (Antonucci et al., 2013; Offer and Fischer, 2018). While informative about relational content, these compositional approaches often overlooked the overarching topological geometry of the ego network—the formal patterns of connectivity, cohesion, and segmentation that govern resource flows, social capital, and information diffusion among alters and between ego and alters. When structural signatures were used, they were often limited to basic network statistics such as size and clustering (Burt, 1992; Marsden, 1987). More recent work has gone beyond this approach, using a richer set of structural signatures, built from alter-to-alter patterns of connectivity, to develop data-driven typologies that classify ego-networks into theoretically meaningful types (Bidart et al., 2018; Giannella and Fischer, 2016; Vacca, 2020).

Our analysis extends previous scholarship in several substantive directions. We begin by mapping the correlation matrix of egocentric graph metrics, showing how subgroup modularity and dyadic density systematically trade off as personal networks expand in volume. To ground these structural signatures in concrete social configurations, we draw on Vacca’s (2020) emphasis on medoid representations to identify empirical archetype exemplars and visualize their relational structures using force-directed network graphs. Moving beyond manual heuristics and algorithmic clustering assignments, we then train an empirical classification decision tree that predicts typology membership with 92.9% accuracy and establishes explicit topological cutoffs. Overcoming the static constraints of cross-sectional surveys, we evaluate semester-to-semester Markov state transitions across 1,900 longitudinal intervals to measure the empirical stability and structural mobility of personal network forms over time. Finally, we bridge the longstanding divide between structural and compositional network traditions by examining functional social support provisions, revealing how topological structure governs relational multiplexity and emotional bandwidth.

## 2 Recent Developments in Personal Network Typologies

The ambition to classify personal networks into discrete structural types represents an enduring program within sociocentric and egocentric analysis. Rather than treating personal

networks as undifferentiated aggregations of ties, typology research seeks to uncover recurrent configurations of interpersonal relations that reflect fundamental principles of social organization (Fischer, 1982; McCarty, 2002; Perry et al., 2018). Over the past two decades, this literature has progressed across three interrelated currents: (1) the transition from attribute-based compositional profiles to purely topological graph structures; (2) the debate between deductive theoretical archetypes and inductive algorithmic clustering; and (3) alternative frameworks centered on structural cohesion, fragmentation, and hierarchical deconstruction.

## 2.1 The Compositional vs. Structural Divide in Network Typologizing

Early typological research predominantly focused on network composition—the demographic attributes, role categories, and institutional contexts characterizing an individual’s contacts (Antonucci et al., 2013; Offer and Fischer, 2018). In a recent influential contribution in this tradition, Giannella and Fischer (2016) used Random Forests on detailed survey data from Northern California ( $N = 1,050$ ) to derive an inductive typology of egocentric networks. Combining over 40 survey descriptors into seven core dimensions (such as non-kin interaction, kin proximity, kin support, church, and work involvement), they reliably placed respondents into seven distinct profiles: “career-and-friends” (24%), “family-and-community” (20%), “family-only” (16%), “untethered” (8%), “energetic” (7%), “withdrawn” (6%), and “home-and-church” (5%).

Subsequent scholarship has extended this compositional paradigm to large national panels and vulnerable populations. Laier et al. (2022) applied the Random Forest framework to the German Socio-Economic Panel (SOEP,  $N = 8,341$ ), identifying fine-grained compositional types based on core discussion networks to show how relational repertoires evolve across the life course. Pelle and Pappadà (2021) developed a distance-based clustering methodology for mixed-type survey data from the Italian National Statistical Institute ( $N = 4,495$ ), grouping elderly individuals living alone into distinct vulnerability profiles based on contact frequency, support type, and kin availability. Extending this logic to romantic dyads, Kennedy et al. (2023) introduced the concept of “duocentric networks” among low-income newlyweds ( $N = 207$ ), clustering couples by spousal network overlap and the balance between family and friend ties to reveal how shared relational ecologies shape marital support.

Yet, as McCarty (2002) argued in an early intervention, compositional summaries treat the personal network as an unordered collection of alters, completely obscuring the structural patterns connecting alters to one another. McCarty showed that eliciting large personal networks (60 alters and 1,770 evaluated pairs) reveals substantial structural heterogeneity in

network density, component counts, and cohesive subgroups that cannot be predicted from ego–alter attributes. Because the topological geometry of alter–alter ties governs resource flows, social capital, normative constraints, and behavioral autonomy (Burt, 1992; Coleman, 1988; Granovetter, 1973), classifying personal networks strictly by their structural topology provides a more direct window into the relational mechanisms that organize social life.

## 2.2 Deductive Theoretical Archetypes vs. Inductive Clustering

The pursuit of purely structural typologies reached a turning point with Bidart et al. (2018), who analyzed longitudinal qualitative and network data from young adults in France ( $N = 87$ ). Rejecting compositional descriptors, Bidart et al. formulated six theoretical archetypes defined solely by alter–alter graph metrics: “Regular Dense” (small, single-clique enclosures), “Centered Dense” (dense cores surrounded by peripheral nodes), “Centered Star” (radial networks dominated by central broker alters), “Segmented” (decentralized, disconnected components), “Pearl Collar” (multiple distinct cliques linked sequentially in a ring or pathway), and “Dispersed” (fragmented, sparse collections of isolates). To assign networks to these archetypes, they proposed a deductive classification tree based on heuristic cutoff values for alter–alter density, Freeman betweenness centralization ( $> 0.20$ ), diameter, and component shares. While theoretically compelling, Bidart et al.’s framework relied on subjective cutoffs derived from a modest sample, raising questions about whether their archetypes reflected universal structural forms or idiosyncratic artifacts of their analytical rules.

To evaluate this question systematically, Vacca (2020) conducted a comparative investigation across six diverse cross-sectional datasets ( $N = 1,460$ ), encompassing immigrants in Southern Europe, disaster survivors in Florida and Ecuador, residents of segregated neighborhoods, and a representative Bay Area sample. Vacca developed an inductive community-detection method using Girvan–Newman modularity partitioning to summarize personal network structure through three properties: the number of cohesive subgroups ( $\geq 3$  nodes), the number of isolated dyads/singletons, and partition modularity. By applying  $k$ -medoids clustering to these metrics, Vacca showed that personal network structures naturally coalesce into distinct inductive groups. Crucially, Vacca revealed substantial discordance and cross-classification between Bidart et al.’s deductive assignments and inductive cluster solutions. Inductive clustering demonstrated that empirical networks rarely conform cleanly to rigid theoretical boundaries, underscoring the need for data-driven classification methods that capture authentic structural variation.

## 2.3 Cohesion, Fragmentation, and Hierarchical Deconstruction

Parallel to the Bidart–Vacca contributions, a complementary line of scholarship has examined the fundamental dimensions underlying structural variation. In representative urban surveys in Spain ( $N = 403$ ), [Maya-Jariego and Holgado \(2015\)](#) used exploratory factor analysis on density, centralization, clique counts, and components, showing that personal network variability is organized along two primary axes: structural cohesion and fragmentation. Building on this foundation, [Maya-Jariego \(2021\)](#) developed a structural classification based on centralization, the number of cliques, and component counts, identifying four empirical ego-network types: “dense,” “intermediate,” “clustered,” and “fragmented.” These studies showed that individual differences in interpersonal environments are primarily structured by the tension between cohesive solidarity and subgroup fragmentation.

Moving beyond static graph metrics, [Maya-Jariego and González-Tinoco \(2023\)](#) introduced a “hierarchical deconstruction procedure” that evaluates network topology by iteratively eliminating nodes with the highest betweenness centrality. Analyzing longitudinal networks of 69 university students, they found that dense, highly cohesive networks exhibit prolonged resistance to fragmentation, whereas networks organized around brokerage rapidly deconstruct into disjoint components. This iterative deconstruction showed that personal networks possess hierarchical, nested subgroup structures that determine their resilience to disruption.

Most recently, [González-Casado et al. \(2024\)](#) addressed the critique that previous typology studies relied on ad-hoc, arbitrarily selected graph metrics. Analyzing four extensive datasets across Spain and Ecuador, they applied systematic dimensionality reduction (PCA and UMAP) across a comprehensive battery of over 14 topological metrics (including transitivity, path length, degree dispersion, modularity, and centralization). Their findings showed that the structural space of personal networks is overwhelmingly governed by two universal axes: (1) global cohesion (the fundamental mathematical tradeoff between network size and density) and (2) internal structural differentiation (the balance between modular community segregation and centralized brokerage).

## 2.4 Extant Gaps in the Current Literature

Despite these significant methodological advances, the existing literature on personal network typologies exhibits three critical gaps: First, virtually all prior structural typology studies ([González-Casado et al., 2024](#); [Maya-Jariego, 2021](#); [McCarty, 2002](#); [Vacca, 2020](#)) rely strictly on single cross-sectional snapshots. Even longitudinal studies ([Bidart and Lavenu, 2005](#); [Bidart et al., 2018](#); [Maya-Jariego and González-Tinoco, 2023](#)) lacked the sample scale or analytical framework to model state transitions across structural types. Accordingly, whether

personal network types represent permanent individual traits or dynamic developmental regimes through which individuals transition over time remains an open empirical question. Second, methodologically, researchers remain trapped between Bidart et al.’s transparent but arbitrary manual heuristics and Vacca’s or González-Casado et al.’s inductive clustering algorithms, which assign cluster memberships within a given sample as an algorithmic “black box” without providing explicit, portable decision rules that other scholars can readily apply. Finally, typological research has remained almost entirely structural and descriptive, focusing on defining and comparing topological forms without examining how different network structures systematically shape substantive social support, relational multiplexity, and affective tie closeness.

The present study addresses each of these three limitations by leveraging the NetHealth Study’s longitudinal design, institutional setting, and relational depth. First, to overcome the cross-sectional constraint, we analyze eight waves of survey data tracking an undergraduate cohort across their first three collegiate years. This multi-wave design allows us to examine both cumulative relational capital accumulated across college ( $N = 701$ ) and semester-to-semester Markov state transitions ( $N = 1,900$  intervals), directly evaluating the empirical persistence versus structural mobility of personal network regimes over time. Second, to resolve the dilemma between arbitrary heuristics and uninterpretable algorithmic clustering, we train an empirical classification decision tree that reproduces unsupervised cluster assignments with 92.9% accuracy. This model yields explicit, data-driven cutoffs—identifying alter–alter density as the primary structural boundary in dense collegiate environments—offering transparent and portable decision criteria that can be applied in comparative research. Finally, to connect abstract topology to substantive social capital, we evaluate alter-level functional support provision across four key domains (social companionship, informational advice, emotional comfort, and financial assistance) alongside composite measures of support multiplexity, relational closeness, and interpersonal trust ( $N = 580$ ).

### 3 Data and Analytical Sample

The empirical analysis draws on the NetHealth Study (Liu et al., 2018; Sepulvado et al., 2022; Wang et al., 2020), a longitudinal investigation that followed an incoming cohort of undergraduate students over four years of college at the University of Notre Dame. Network surveys were administered online via Qualtrics at the beginning and conclusion of each semester across eight distinct waves between Fall 2015 and Spring 2018. The survey used an open-ended name-generator approach, asking respondents to name up to 20 individuals with whom they communicated or interacted. Beginning in Wave 3, respondents could additionally

retain up to five alters from the preceding wave, allowing for up to twenty-five named alters per wave. Name interpreters recorded demographic and relational traits for each alter, and respondents completed alter–alter matrices indicating which of their nominated alters knew one another.

Across all eight survey waves, participants generated 35,912 ego–alter nominations and 174,748 alter–alter evaluations. To capture the full scope of interpersonal connectivity realized during college, we constructed cumulative undirected ego networks by combining all unique alters and realized alter–alter ties reported by each participant across the observation period. Following established methodological criteria for egocentric graph analysis, we restricted the analytic sample to participants who named at least three alters and who reported at least one realized tie between alters. This filtering eliminated isolated dyads and participants with degenerate alter graphs, yielding a final analytic sample of  $N = 701$  participants with complete network and demographic data.

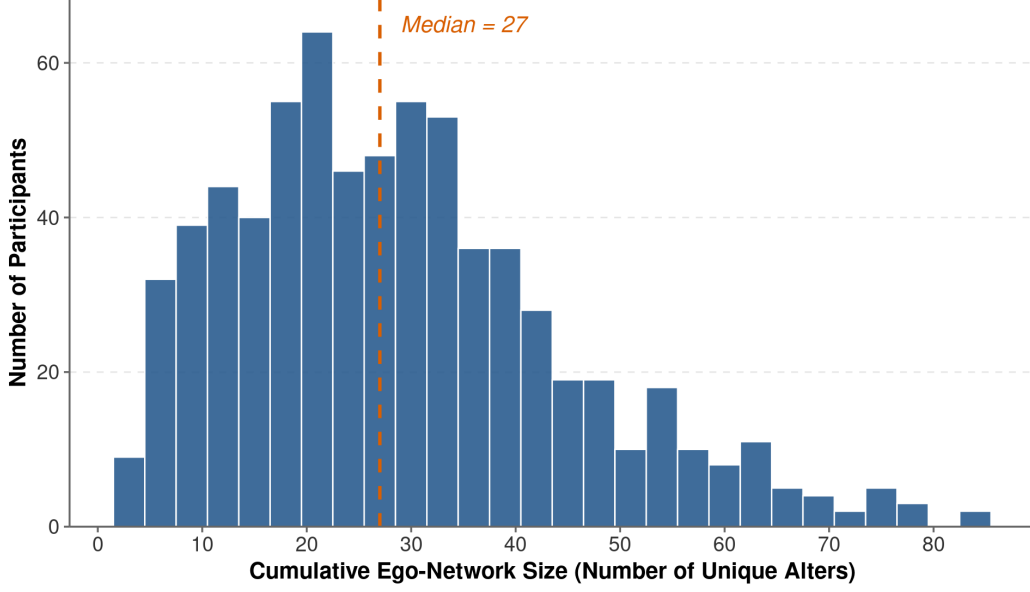
**Table 1:** Descriptive Statistics of Personal Network Structural Metrics by Typology ( $N = 701$ )

| Structural Metric           | Full Sample<br>( $N = 701$ ) | Pearl Collar<br>( $n = 176$ ) | Segmented<br>( $n = 261$ ) | Centered Star<br>( $n = 143$ ) | Regular Dense<br>( $n = 121$ ) |
|-----------------------------|------------------------------|-------------------------------|----------------------------|--------------------------------|--------------------------------|
| Network Size                | 28.8 (16.0)                  | 47.9 (14.0)                   | 27.4 (10.1)                | 21.6 (8.2)                     | 12.6 (6.5)                     |
| Community Modularity        | 0.31 (0.17)                  | 0.46 (0.11)                   | 0.33 (0.13)                | 0.31 (0.11)                    | 0.07 (0.07)                    |
| Network Diameter            | 3.2 (1.3)                    | 4.5 (1.2)                     | 2.7 (0.8)                  | 3.4 (0.9)                      | 1.9 (0.6)                      |
| Betweenness Centralization  | 0.19 (0.15)                  | 0.22 (0.11)                   | 0.11 (0.07)                | 0.40 (0.12)                    | 0.06 (0.07)                    |
| Alter–Alter Density         | 0.38 (0.21)                  | 0.21 (0.06)                   | 0.33 (0.11)                | 0.37 (0.10)                    | 0.77 (0.15)                    |
| Transitivity (Clustering)   | 0.73 (0.15)                  | 0.64 (0.10)                   | 0.74 (0.14)                | 0.68 (0.13)                    | 0.88 (0.09)                    |
| Largest Component Share (%) | 87.0% (18.7)                 | 85.6% (17.5)                  | 77.3% (22.3)               | 97.8% (5.1)                    | 97.2% (6.3)                    |

*Note:* Standard deviations are reported in parentheses. Size represents the count of unique alters. Centralization refers to Freeman alter–alter betweenness centralization. Modularity is computed via the Louvain community detection algorithm on alter–alter ties. Largest component share indicates the proportion of alters belonging to the largest connected component.

Table 1 reports descriptive statistics for the personal network structural metrics across the full analytic sample and within each extracted typology. Figure 1 shows the distribution of cumulative ego-network size across the study cohort. The empirical spread underscores that static single-wave snapshots substantially underestimate the total volume of social ties maintained by young adults in residential campus environments. The right tail extends beyond 80 cumulative contacts, with an average of 28.8 alters ( $SD = 16.0$ ) and a median of 26 alters, highlighting the capacity of highly engaged individuals to sustain extensive interpersonal environments over time.





**Figure 1:** Cumulative Ego-Network Size Distribution Across NetHealth Participants

*Note:* Distribution of unique alters nominated across Waves 1 through 8 for analytic-sample participants ( $N = 701$ ). The dashed vertical line indicates the sample median of 26 alters; the solid curve represents kernel density smoothing.

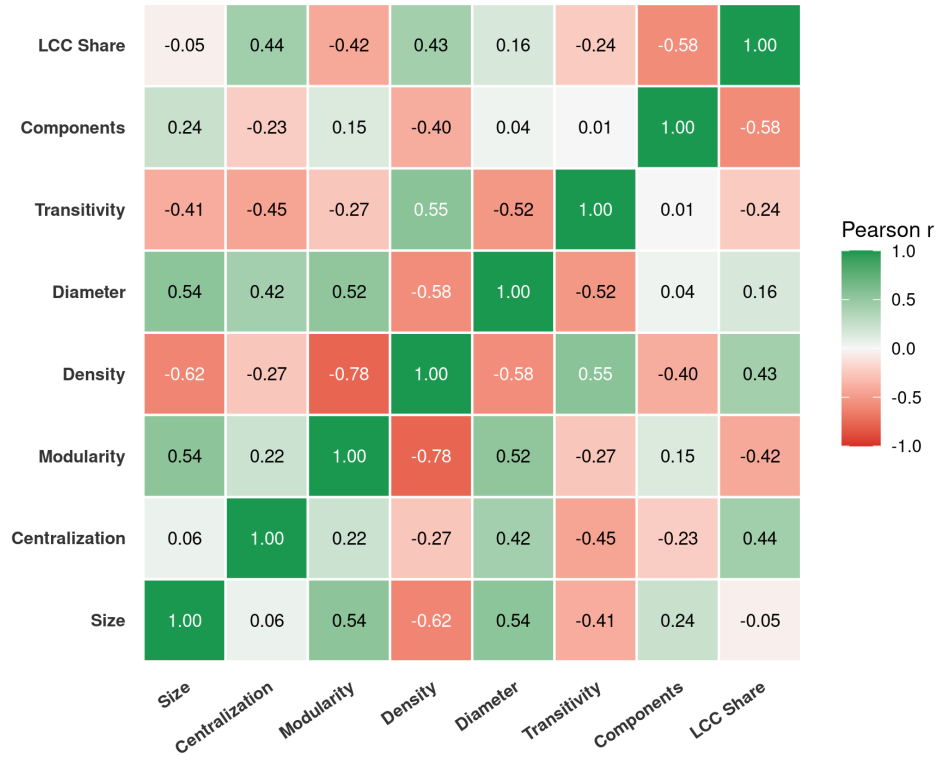
## 4 Results

### 4.1 Structural Network Typologies

To classify personal networks into structural types without imposing *ex ante* categories, we standardized five core graph metrics ( $z$ -scores): network size ( $n_i$ ), dyadic density ( $d_i$ ), Louvain community modularity ( $Q_i$ ), graph diameter of the largest connected component ( $D_i$ ), and Freeman alter–alter betweenness centralization ( $C_{B,i}$ ). Formally, each ego network  $i$  is represented by a standardized feature vector  $\mathbf{z}_i \in \mathbb{R}^5$ :

$$\mathbf{z}_i = \left( \frac{n_i - \bar{n}}{\text{SD}(n)}, \frac{d_i - \bar{d}}{\text{SD}(d)}, \frac{Q_i - \bar{Q}}{\text{SD}(Q)}, \frac{D_i - \bar{D}}{\text{SD}(D)}, \frac{C_{B,i} - \bar{C}_B}{\text{SD}(C_B)} \right)^\top \quad (1)$$

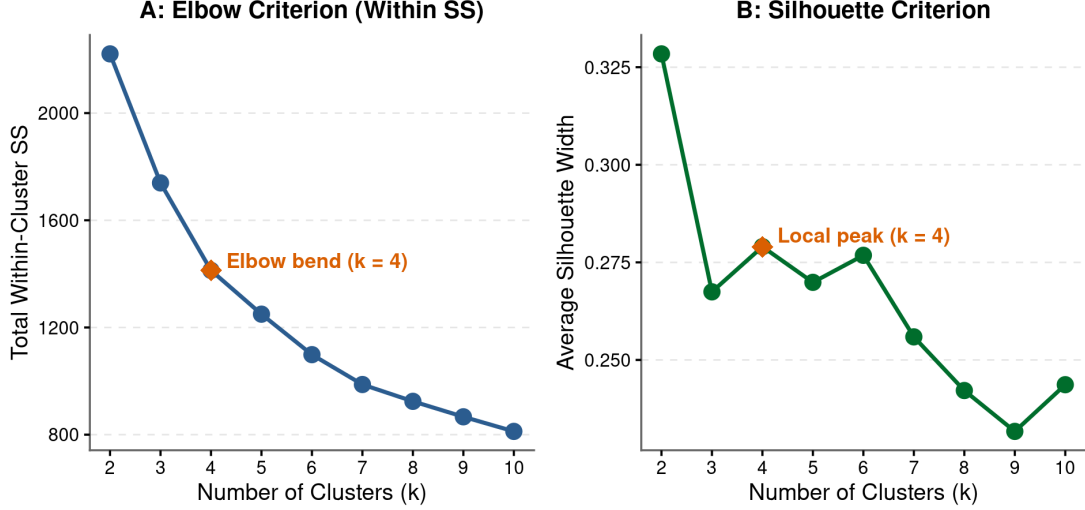
The correlation matrix of these ego-network features is shown in Figure 2. Alter–alter density exhibits strong negative correlations with both network size ( $r = -0.58$ ) and Louvain modularity ( $r = -0.66$ ). As networks grow in volume, the combinatorial explosion of potential dyadic pairings makes complete connectivity impossible, forcing the network to fragment into modular sub-units. Conversely, network size correlates positively with modularity ( $r = +0.50$ ), graph diameter ( $r = +0.48$ ), and the proportion of alters in the largest connected component ( $r = +0.42$ ). Global clustering (transitivity) remains consistently high across the sample



**Figure 2:** Pairwise Pearson Correlation Heatmap Across Personal Network Metrics

*Note:* Pairwise Pearson correlation coefficients among eight structural ego-network metrics ( $N = 701$ ). Green indicates positive association; red indicates negative association.

but correlates positively with density ( $r = +0.49$ ) and negatively with size ( $r = -0.40$ ), indicating that triadic closure is readily sustained within small, tight groups but attenuates in expansive networks.



**Figure 3:** Cluster Partition Quality Diagnostics Across Candidate Solutions ( $k = 2$  to  $10$ )

*Note:* Panel A displays the total within-cluster sum of squares (elbow curve) across cluster solutions from  $k = 2$  to  $10$ , with the elbow inflection highlighted at  $k = 4$ . Panel B displays the average silhouette width, with a distinct local peak at  $k = 4$  (0.279).

Using these standardized features, we performed unsupervised  $k$ -means clustering across candidate values of  $k \in \{2, \dots, 10\}$  by minimizing the total within-cluster sum of squares:

$$\min_{\mathcal{S}=\{S_1, \dots, S_k\}} \sum_{c=1}^k \sum_{\mathbf{z}_i \in S_c} \|\mathbf{z}_i - \boldsymbol{\mu}_c\|^2 \quad (2)$$

Where  $\boldsymbol{\mu}_c$  denotes the centroid of cluster  $S_c$ . We evaluated partition quality using silhouette coefficients, elbow plots of within-cluster variance, and the Calinski–Harabasz index. As shown in Figure 3 and Table 2, the four-cluster partition achieved the most parsimonious and substantively interpretable grouping. Panel A of Figure 3 displays the elbow criterion, showing a pronounced bend at  $k = 4$  (capturing 59.6% of total variance), after which the curve flattens with diminishing marginal returns. Panel B displays the average silhouette width across candidate partitions, revealing a distinct local peak at  $k = 4$  (0.279), higher than at  $k = 3$  (0.267) or  $k = 5$  (0.270). Together, the elbow inflection and silhouette criterion confirm that the four-cluster solution represents the optimal and most reliable partition for these data.

As shown in Table 1, the four clusters correspond to the primary topological configurations identified by Bidart et al. (2018):

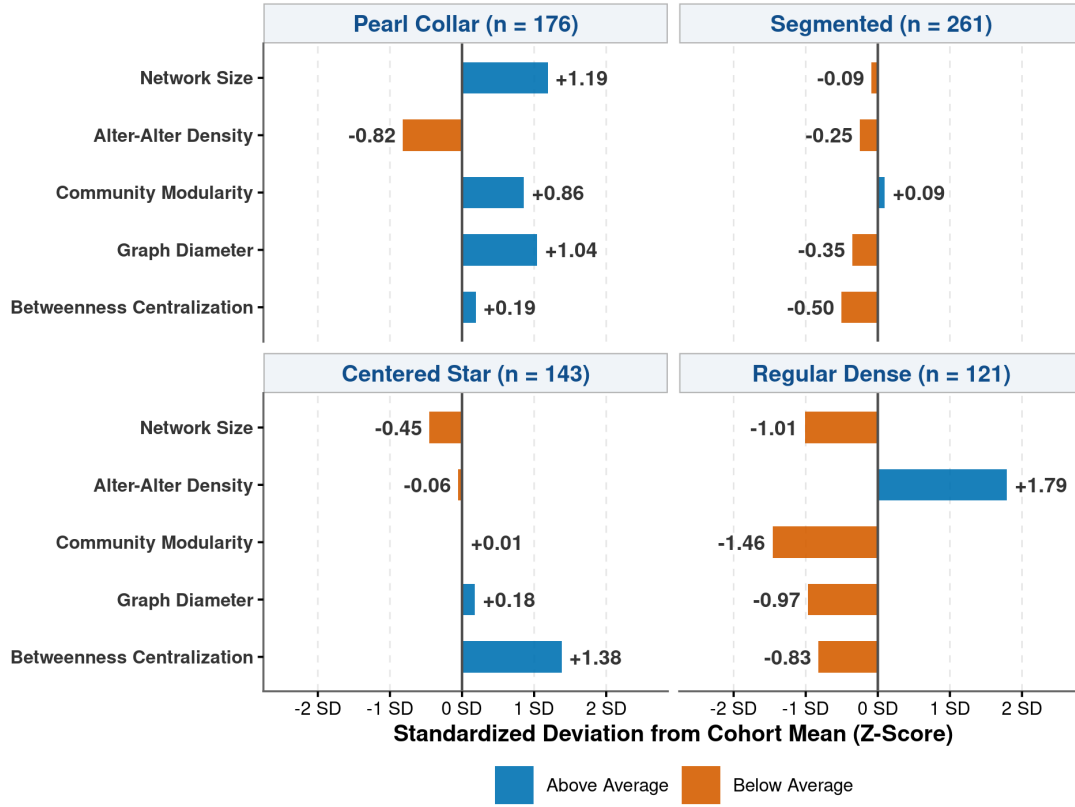
**Table 2:** Cluster Partition Quality Diagnostics Across Candidate Solutions ( $k = 2$  to 10)

| Clusters<br>( $k$ ) | Within-Cluster<br>SS | Variance<br>Explained (%) | Average<br>Silhouette | Calinski–Harabasz<br>Index |
|---------------------|----------------------|---------------------------|-----------------------|----------------------------|
| 2                   | 2220.9               | 36.5%                     | 0.328                 | 402.6                      |
| 3                   | 1739.5               | 50.3%                     | 0.267                 | 353.2                      |
| <b>4</b>            | <b>1413.1</b>        | <b>59.6%</b>              | <b>0.279</b>          | <b>343.1</b>               |
| 5                   | 1249.6               | 64.3%                     | 0.270                 | 313.3                      |
| 6                   | 1098.6               | 68.6%                     | 0.277                 | 303.8                      |
| 7                   | 987.1                | 71.8%                     | 0.256                 | 294.4                      |
| 8                   | 924.4                | 73.6%                     | 0.242                 | 275.8                      |
| 9                   | 866.7                | 75.2%                     | 0.232                 | 262.8                      |
| 10                  | 812.2                | 76.8%                     | 0.244                 | 254.1                      |

*Note:* Diagnostics evaluated across standardized structural metrics ( $z$ -scores of size, modularity, diameter, betweenness centralization, and density) for  $N = 701$  participants. Bold row highlights the optimal four-cluster solution, which maximizes the local silhouette width while reaching the elbow inflection.

1. **Pearl Collar Networks** ( $n = 176$ , 25.1%): Characterized by expansive network size ( $\bar{x} = 47.9$ ), the highest graph diameter ( $\bar{x} = 4.50$ ), elevated modularity ( $\bar{x} = 0.456$ ), low density ( $\bar{x} = 0.207$ ), and moderate betweenness centralization ( $\bar{x} = 0.218$ ). These networks consist of multiple distinct cliques linked in an elongated chain or ring by bridging individuals.
2. **Segmented Networks** ( $n = 261$ , 37.2%): Comprises moderately sized networks ( $\bar{x} = 27.4$ ) exhibiting high modularity ( $\bar{x} = 0.327$ ), moderate diameter ( $\bar{x} = 2.70$ ), low centralization ( $\bar{x} = 0.112$ ), and moderate density ( $\bar{x} = 0.330$ ). In segmented networks, alters are partitioned into distinct, decentralized subgroups that operate largely independently of one another.
3. **Centered Star Networks** ( $n = 143$ , 20.4%): Distinguished by exceptionally high betweenness centralization ( $\bar{x} = 0.399$ ), elevated diameter ( $\bar{x} = 3.38$ ), moderate modularity ( $\bar{x} = 0.313$ ), and moderate density ( $\bar{x} = 0.370$ ). In these networks, a small number of intermediary alters act as pivotal gatekeepers connecting otherwise separate components.
4. **Regular Dense Networks** ( $n = 121$ , 17.3%): Characterized by small cumulative size ( $\bar{x} = 12.6$ ), low diameter ( $\bar{x} = 1.90$ ), negligible modularity ( $\bar{x} = 0.065$ ), low centralization ( $\bar{x} = 0.063$ ), and very high alter–alter density ( $\bar{x} = 0.765$ ). In regular dense networks, nearly every alter knows every other alter, forming a tightly cohesive, redundant social enclave. Notably, we find no empirical support for separate “Centered Dense” or “Dispersed” clusters in this sample, indicating that in residential university

contexts, high density and high centralization rarely co-occur.



**Figure 4:** Standardized Topological Profiles Across Personal Network Typologies

*Note:* Profile panels display standardized deviations ( $z$ -scores) from the cohort mean across the five core structural dimensions for each personal network typology ( $N = 701$ ). Blue bars indicate metrics above the sample average; vermilion bars indicate metrics below it.

Figure 4 displays the standardized topological signatures ( $z$ -scores) across the five core structural metrics for each personal network typology, illustrating how each configuration departs from the cohort mean. In the top-left panel, the *Pearl Collar* configuration ( $n = 176$ ) is defined by four above-average metrics: expansive network size (+1.19 SD), the highest graph diameter in the cohort (+1.04 SD), elevated community modularity (+0.86 SD), and slightly elevated betweenness centralization (+0.19 SD). These positive features are counterbalanced by a single, pronounced negative dimension: density ( $-0.82$  SD), capturing the sequential chaining of distinct modular cliques across bridging pathways.

In the top-right panel, the *Segmented* configuration ( $n = 261$ ) is characterized by a decentralized, compact profile in which four of the five dimensions fall below the average. Its primary defining feature is low betweenness centralization ( $-0.50$  SD), accompanied by below-average graph diameter ( $-0.35$  SD) and below-average density ( $-0.25$  SD). Network size sits close to the cohort mean ( $-0.09$  SD), while community modularity is slightly positive (+0.09 SD).

SD), indicating that segmentation in these networks reflects localized subgroup autonomy rather than chain-like modular bridging.

In the bottom-left panel, the Centered Star configuration ( $n = 143$ ) is dominated by a substantial spike in betweenness centralization (+1.38 SD)—the largest positive deviation observed for any metric outside of density for Regular Dense networks. Crucially, this focal brokerage pattern operates within relatively compact networks: network size is distinctly below average ( $-0.45$  SD;  $\bar{x} = 21.6$  alters), while density ( $-0.06$  SD) and community modularity ( $+0.01$  SD) hover near the sample mean, and diameter is modestly elevated ( $+0.18$  SD).

Finally, in the bottom-right panel, the Regular Dense configuration ( $n = 121$ ) exhibits an inverse structural profile to the Centered Star network. It is anchored by extreme density (+1.79 SD), while all other four structural dimensions are strongly below average: modularity ( $-1.46$  SD), network size ( $-1.01$  SD), graph diameter ( $-0.97$  SD), and betweenness centralization ( $-0.83$  SD). In this regime, interpersonal ties coalesce into small, tightly knit, and redundant social enclaves where virtually all alters are directly connected.

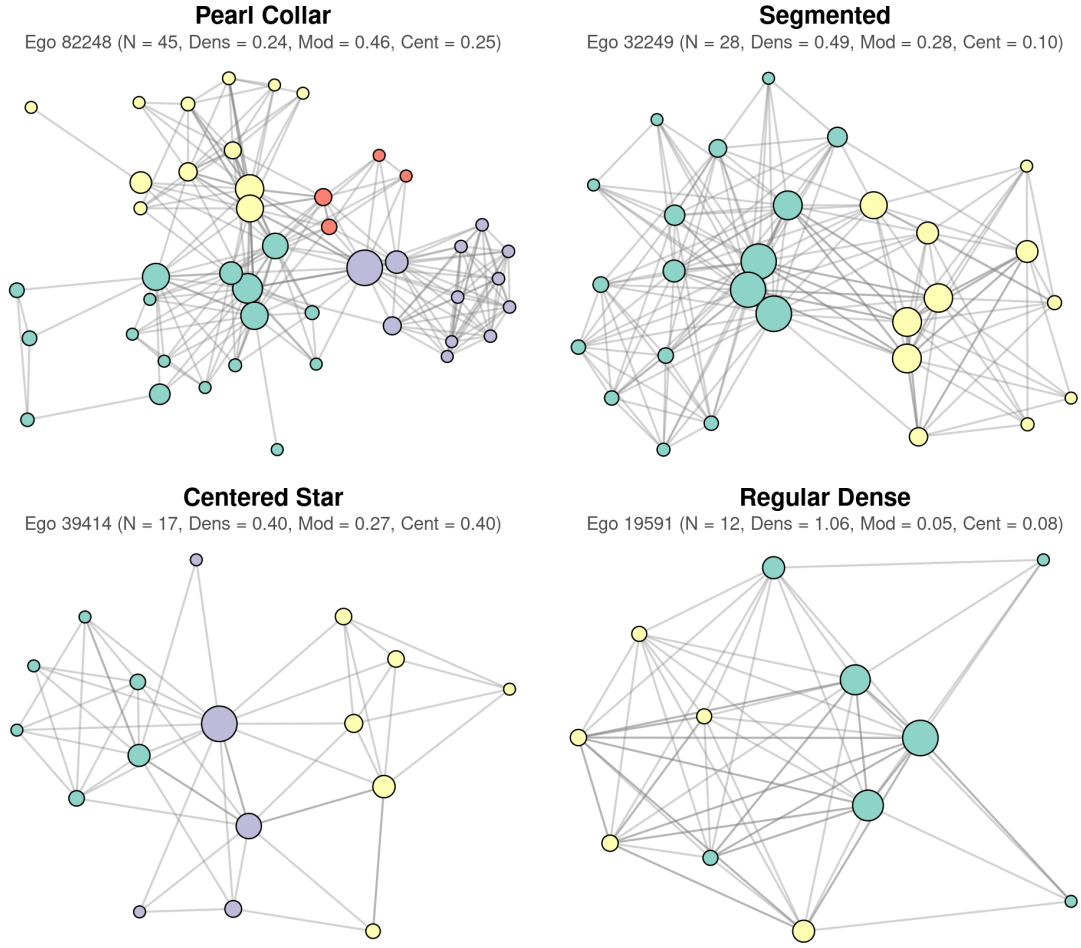
## 4.2 Archetype Network Exemplars: Empirical Medoids

To ground the four-fold typology more concretely, we identified the empirical medoid ego for each cluster—namely, the individual whose standardized structural coordinates minimize Euclidean distance to all other members in that cluster:

$$i_c^* = \arg \min_{i \in S_c} \sum_{j \in S_c} \|\mathbf{z}_i - \mathbf{z}_j\|_2 \quad (3)$$

Figure 5 presents force-directed network graph layouts of these four empirical archetypes, with nodes sized by betweenness centrality and colored by Louvain community membership.

Figure 5 illustrates the structural signatures defining each network regime. The Pearl Collar medoid (Ego 82248;  $N = 47$ , diameter = 5, modularity = 0.45) displays a large, elongated configuration in which multiple distinct modules are threaded together sequentially via bridging ties. The Segmented medoid (Ego 32249;  $N = 30$ , modularity = 0.30) exhibits clear multi-colored community clustering without a dominant central gatekeeper. By contrast, the Centered Star medoid (Ego 39414;  $N = 17$ , centralization = 0.42) features prominent broker nodes that anchor the entire network and funnel communication between otherwise disconnected alters. Finally, the Regular Dense medoid (Ego 19591;  $N = 12$ , density = 0.79) forms a single, tightly bound cluster in which nearly all potential ties are realized and betweenness centrality is uniformly low.

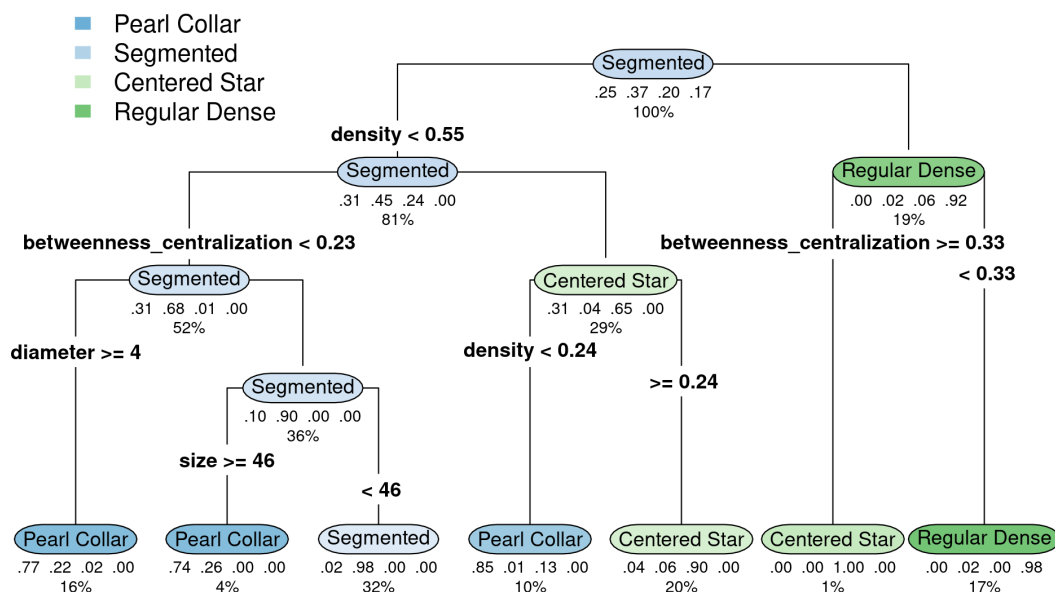


**Figure 5:** Empirical Network Archetypes: Force-Directed Layouts of Cluster Medoids

*Note:* Force-directed network graphs (**ggraph** stress layout) of the four empirical cluster medoids. Nodes represent nominated alters; edges represent reported alter-alter ties. Node size reflects alter betweenness centrality; node color reflects Louvain community assignment.

### 4.3 Empirical Decision Trees: Data-Driven Cutoffs vs. Heuristic Frameworks

A central objective of this article is to move beyond subjective, manual heuristics and establish data-driven classification rules. To that end, we estimated a recursive partitioning classification tree (Breiman et al., 1984) in R using the `rpart` package to predict cluster membership based on structural properties. To evaluate how institutional context alters structural thresholds, Figure 7 contrasts Bidart et al.’s (2018) theoretical tree derived from French young adults against our empirical NetHealth classification tree.

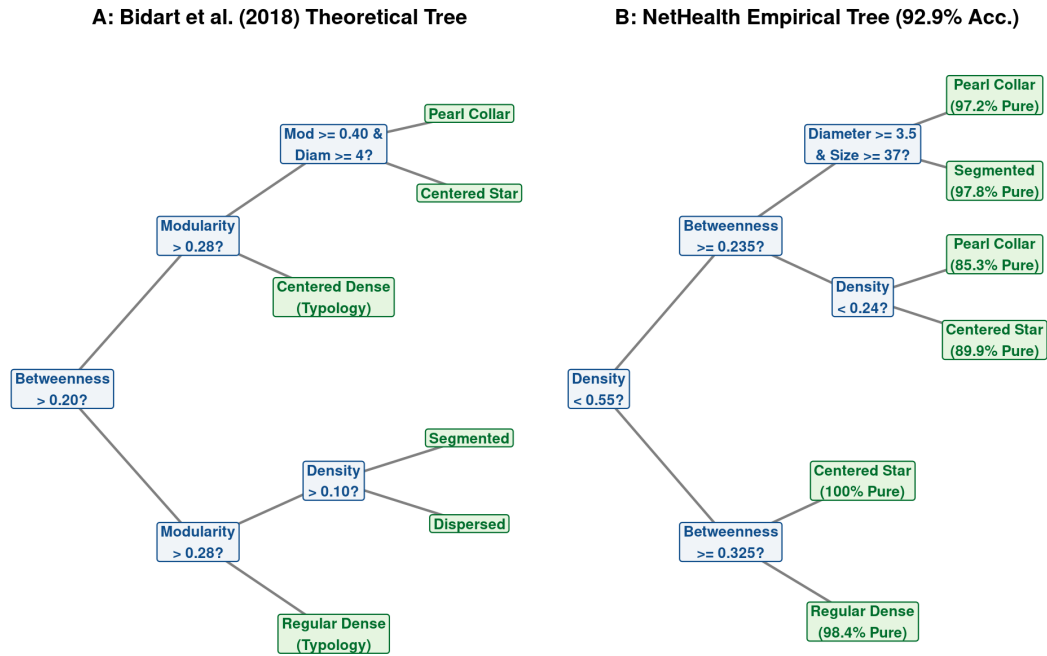


**Figure 6:** Empirical Decision Tree for Classifying Personal Network Typologies

*Note:* Classification decision tree (`rpart`) predicting personal network typologies from structural properties ( $N = 701$ ). Terminal leaves display predicted class, classification accuracy, and sample share. Overall classification accuracy is 92.9%.

As shown in Figure 6 and Figure 7, our empirical decision tree achieves 92.9% overall accuracy and reveals critical differences from earlier heuristic frameworks. Whereas Bidart et al. (2018) initiated their classification tree on betweenness centralization ( $> 0.20$ ), the empirical *NetHealth* tree identifies density as the primary root split at a cutoff of 0.55: networks with density exceeding 0.55 and low centralization ( $< 0.33$ ) are classified as Regular Dense with 98.4% accuracy. For lower-density networks, the algorithm splits on betweenness centralization at 0.24, isolating Centered Stars (89.9% classification accuracy), followed by diameter ( $\geq 3.5$ ) and size ( $\geq 37$ ) to delineate Pearl Collar networks from Segmented networks





**Figure 7:** Comparative Decision Tree Graphic: Bidart Theoretical Heuristics vs. NetHealth Empirical Cutoffs

*Note:* Side-by-side comparison of classification decision trees. Panel A displays Bidart et al.’s (2018) theoretical heuristics based on French young adults. Panel B displays the empirical NetHealth classification tree for U.S. college students (92.9% accuracy).

(97.8% classification accuracy). These cutoffs demonstrate that the high-contact residential campus environment substantially shifts baseline density upward relative to the general young adult population.

#### 4.4 Longitudinal Trajectory Transitions and Typology Dynamics

While cumulative networks capture total relational capital, egocentric environments evolve dynamically over time (Bidart and Lavenu, 2005). To investigate stability and mobility across typologies, we extracted network metrics wave-by-wave across all eight survey administrations ( $N = 2,821$  ego-wave observations) and classified each observation using our empirical decision rules. We observed  $N = 1,900$  adjacent semester-to-semester transitions ( $t \rightarrow t + 1$ ). The discrete-time Markov chain is parameterized by the transition probability matrix  $\mathbf{P} = [P_{ij}]$ :

$$P_{ij} = \Pr(S_{t+1} = j \mid S_t = i), \quad \sum_{j=1}^4 P_{ij} = 1 \quad (4)$$

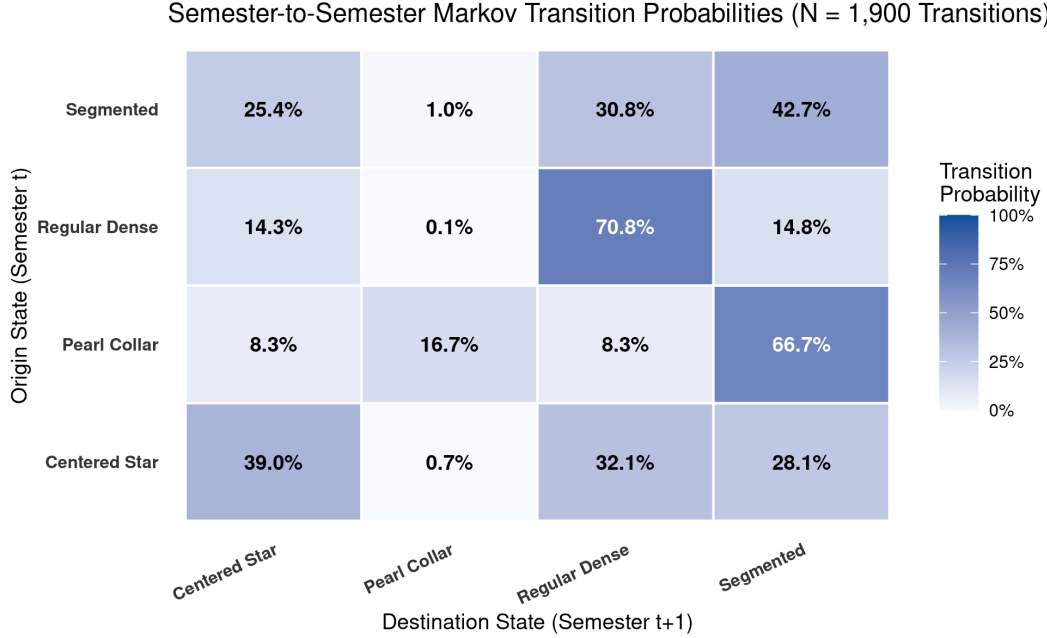
where  $i, j \in \{\text{Centered Star, Pearl Collar, Regular Dense, Segmented}\}$ . Table 3 and Figure 8 display the resulting Markov transition probability matrix.

**Table 3:** Semester-to-Semester Markov State Transition Probability Matrix ( $N = 1,900$  Transitions)

| Origin State<br>(Semester $t$ ) | Destination State (Semester $t + 1$ ) |              |               |           | Total<br>Transitions |
|---------------------------------|---------------------------------------|--------------|---------------|-----------|----------------------|
|                                 | Centered Star                         | Pearl Collar | Regular Dense | Segmented |                      |
| Centered Star                   | 39.0%                                 | 0.7%         | 32.1%         | 28.1%     | 420                  |
| Pearl Collar                    | 8.3%                                  | 16.7%        | 8.3%          | 66.7%     | 12                   |
| Regular Dense                   | 14.3%                                 | 0.1%         | 70.8%         | 14.8%     | 965                  |
| Segmented                       | 25.4%                                 | 1.0%         | 30.8%         | 42.7%     | 503                  |

*Note:* Transition probabilities based on  $N = 1,900$  observed semester-to-semester transitions across Waves 1 through 8. Row percentages sum to 100%. Diagonal entries reflect structural state persistence across semesters.

Figure 8 and Table 3 reveal striking differences in structural persistence across typologies. The Regular Dense state exhibits the highest stability, with a 70.8% probability of remaining in that same state in the subsequent semester. By contrast, Segmented and Centered Star networks display moderate persistence (42.7% and 39.0%, respectively), with substantial transition flows between one another and into the dense cluster, perhaps driven by gaining or losing ties to high-betweenness alters. Pearl Collar networks are comparatively transient in single-wave snapshots (16.7% persistence), functioning primarily as transient structures that coalesce as students weave together contacts accumulated across different facets of campus life.



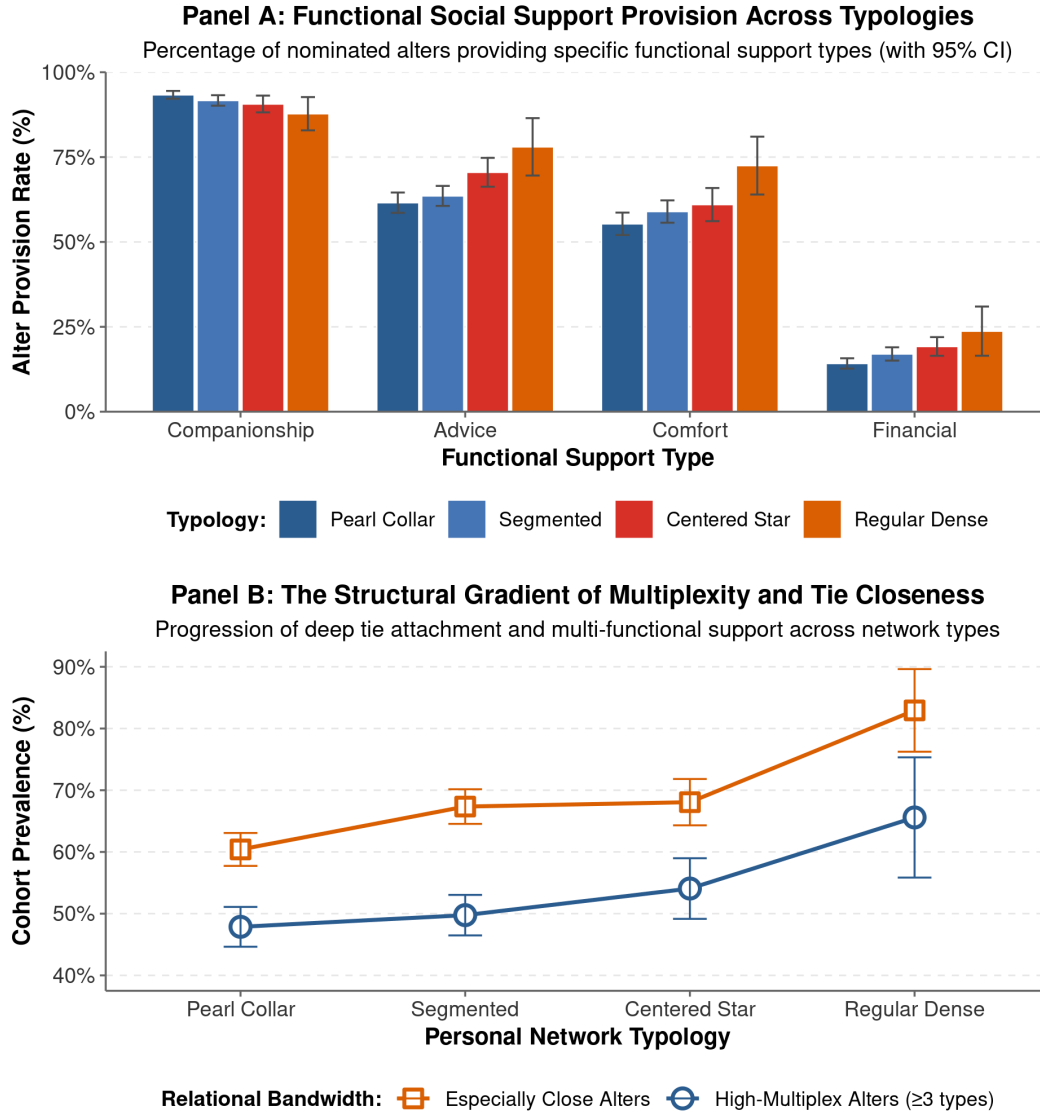
**Figure 8:** Semester-to-Semester Markov State Transition Probability Heatmap

*Note:* Transition probability matrix across personal network states for  $N = 1,900$  longitudinal intervals. Cell values indicate the empirical probability of transitioning from the origin state at semester  $t$  to the destination state at semester  $t + 1$ .

## 4.5 Functional Social Support and Relational Multiplexity Across Typologies

As we noted at the outset, a persistent divide in ego network research separates the compositional tradition—which focuses on relational content, functional aid, and social support (Giannella and Fischer, 2016; Laier et al., 2022; Pelle and Pappadà, 2021)—from the purely structural tradition (Bidart et al., 2018; González-Casado et al., 2024; McCarty, 2002; Vacca, 2020). By examining alter-level support evaluations in our analytic sample ( $N = 580$  participants with complete support records), we directly address this gap by assessing whether distinct topological configurations systematically shape relational multiplexity and functional support bandwidth. Table 4 reports descriptive statistics and analysis of variance across network typologies for eight support and relationship characteristics. Figure 9 visualizes these functional support profiles and the structural multiplexity gradient.

Table 4 and Figure 9 show that while social companionship (hanging out) represents a universal baseline of collegiate sociability ( $\sim 88\%$  to  $93\%$  across all configurations), substantive functional support exhibits a pronounced, monotonic structural gradient across typologies. Informational advice rises systematically from the expansive Pearl Collar configuration ( $61.6\%$ ) and Segmented networks ( $63.6\%$ ) to Centered Stars ( $70.5\%$ ) and reaches its peak



**Figure 9: Functional Social Support Profiles and Multiplexity Gradient Across Personal Network Typologies**

*Note:* Panel A displays the percentage of nominated alters providing specific functional support types across personal network typologies with 95% confidence interval error bars. Panel B displays the monotonic progression of high-multiplex alters ( $\geq 3$  support types) and tie closeness (% especially close alters) across network configurations ( $N = 580$ ).

**Table 4:** Social Support Provision and Functional Multiplexity Across Personal Network Typologies ( $N = 580$ )

| Functional Support Dimension          | Full Sample<br>( $N = 580$ ) | Pearl Collar<br>( $n = 176$ ) | Segmented<br>( $n = 242$ ) | Centered Star<br>( $n = 119$ ) | Regular Dense<br>( $n = 43$ ) | $F$ -Stat | $p$ -value |
|---------------------------------------|------------------------------|-------------------------------|----------------------------|--------------------------------|-------------------------------|-----------|------------|
| Social Companionship (%)              | 91.7% (11.9)                 | 93.3% (7.8)                   | 91.7% (12.3)               | 90.6% (13.8)                   | 87.8% (16.4)                  | 3.03      | 0.029      |
| Informational Advice (%)              | 65.5% (23.4)                 | 61.6% (20.2)                  | 63.6% (23.4)               | 70.5% (23.7)                   | 78.0% (28.3)                  | 8.47      | < 0.001    |
| Emotional Comfort (%)                 | 59.3% (25.7)                 | 55.3% (22.3)                  | 59.0% (26.1)               | 61.0% (27.2)                   | 72.5% (28.5)                  | 5.46      | 0.001      |
| Financial Support (%)                 | 17.1% (15.2)                 | 14.2% (10.3)                  | 17.0% (15.5)               | 19.2% (15.3)                   | 23.7% (24.2)                  | 5.79      | < 0.001    |
| Support Multiplexity Index (0–4)      | 2.34 (0.54)                  | 2.24 (0.45)                   | 2.31 (0.54)                | 2.41 (0.56)                    | 2.62 (0.63)                   | 6.90      | < 0.001    |
| High-Multiplex Alters ( $\geq 3$ , %) | 51.2% (26.0)                 | 47.9% (21.8)                  | 49.8% (26.1)               | 54.1% (27.3)                   | 65.6% (32.6)                  | 6.24      | < 0.001    |
| Tie Closeness (% Especially Close)    | 66.5% (21.5)                 | 60.4% (18.0)                  | 67.4% (22.3)               | 68.1% (20.9)                   | 82.9% (22.4)                  | 14.35     | < 0.001    |
| Alter Trust Rating (1–10 scale)       | 8.68 (0.91)                  | 8.50 (0.94)                   | 8.76 (0.90)                | 8.64 (0.87)                    | 9.02 (0.83)                   | 5.28      | 0.001      |

*Note:* Sample restricted to  $N = 580$  participants with complete alter support evaluations across Waves 2 through 8. Standard deviations are reported in parentheses. Support Multiplexity Index reflects the average count of functional support types (social companionship, advice, emotional comfort, financial assistance) provided per alter (0 to 4 scale). High-multiplex alters represent the percentage of alters providing three or more distinct support functions.  $F$ -statistics and  $p$ -values are derived from one-way analysis of variance across network typologies.

in Regular Dense networks (78.0%;  $F = 8.47, p < 0.001$ ). Emotional comfort exhibits an identical progression, rising from 55.3% in Pearl Collar networks to 72.5% in Regular Dense networks ( $F = 5.46, p = 0.001$ ), while financial assistance displays a matching concentration (14.2% in Pearl Collar vs. 23.7% in Regular Dense;  $F = 5.79, p < 0.001$ ).

Crucially, this functional gradient reflects a fundamental structural tradeoff between topological reach and relational bandwidth. As shown in Panel B of Figure 9, the Support Multiplexity Index—measuring the average number of functional supports provided per alter—increases monotonically from 2.24 in Pearl Collar networks to 2.62 in Regular Dense networks ( $F = 6.90, p < 0.001$ ). Similarly, the share of “high-multiplex” alters providing three or more distinct forms of support rises from 47.9% to 65.6% ( $F = 6.24, p < 0.001$ ). This functional concentration is underpinned by emotional intimacy: alters classified as “especially close” account for only 60.4% of contacts in Pearl Collar networks, but surge to 82.9% in Regular Dense networks ( $F = 14.35, p < 0.001$ ), accompanied by elevated interpersonal trust ( $F = 5.28, p = 0.001$ ). Expansive, chained network configurations like the Pearl Collar maximize structural breadth and bridge across modular student worlds, but they do so by diluting the proportion of multiplex, emotionally intensive, and financially supportive ties. Conversely, small, cohesive cliques sacrifice structural reach and external bridging in order to maximize dense mutual trust, emotional solidarity, and multi-functional safety nets.

## 5 Discussion and Conclusion

### 5.1 Summary of Key Results

Taken together, the empirical findings and topological analyses presented across Tables 1 to 4 and Figures 1 to 9 provide a comprehensive structural cartography of egocentric relational environments in a residential collegiate setting. Rather than supporting a model of continuous, unstructured variation in personal network connectivity, the evidence demonstrates that interpersonal networks naturally coalesce into discrete, recurrent topological configurations that mirror and extend the foundational archetypes identified by Bidart et al. (2018) and Vacca (2020).

At the core of this typology are four distinct structural regimes: expansive and modular “Pearl Collar Networks” ( $\bar{x} = 47.9$  alters), decentralized “Segmented Networks” ( $\bar{x} = 27.4$  alters), centralized “Centered Star Networks” ( $\bar{x} = 21.6$  alters), and tightly cohesive “Regular Dense Networks” ( $\bar{x} = 12.6$  alters). Notably, we find no empirical evidence in this residential context for separate “Centered Dense” or “Dispersed” clusters, indicating that under conditions of daily campus co-presence, extreme density and high betweenness centralization rarely co-occur, while isolates and fragmented dyads are readily integrated into larger relational structures.

Our empirical decision tree moves beyond the subjective heuristics of prior work, classifying network typology membership with 92.9% overall accuracy. The tree establishes portable, data-driven cutoffs, identifying alter–alter density ( $\geq 0.55$ ) as the primary structural threshold separating cohesive social enclaves from internally differentiated configurations, followed by betweenness centralization (0.24), diameter ( $\geq 3.5$ ), and network size ( $\geq 37$ ). These explicit rules provide comparative researchers with objective criteria to classify personal network forms without relying on algorithmic black-box clustering.

Furthermore, our longitudinal Markov transition models reveal that personal network typologies are developmental regimes through which individuals move over time rather than immutable individual traits. While small, cohesive Regular Dense networks exhibit remarkable semester-to-semester persistence (70.8%), students experience substantial structural mobility across the remaining configurations. Expansive Pearl Collar topologies, in particular, appear primarily as cumulative multi-epoch structures that emerge as students weave together contacts across disparate academic, residential, and extracurricular spheres.

Finally, our alter-level functional support analysis bridges the longstanding divide between structural and compositional network paradigms. While baseline companionship is pervasive across all typologies ( $\sim 88\%$  to  $93\%$ ), substantive functional aid (informational advice, emotional comfort, financial assistance), support multiplexity ( $\geq 3$  functions), tie closeness,

and interpersonal trust concentrate heavily in small, tightly knit Regular Dense enclaves. Expansive Pearl Collar networks maximize topological reach and bridge across modular student worlds, but do so by diluting the proportion of multiplex, emotionally intensive, and high-trust ties.

## 5.2 Limitations and Suggestions for Future Work

Several methodological and contextual limitations of the present study warrant consideration and point to fertile avenues for future research.

First, the empirical setting is restricted to undergraduate students embedded within a full-time, highly residential university campus. While this collegiate environment provides an unparalleled observational setting for tracking dense interpersonal interactions, alter–alter connectivity, and multiplex tie formation, the baseline network size and density thresholds reflect the spatial and social opportunities of residential student life. In populations facing different organizational foci—such as older adults living alone, working-class households, or geographically dispersed immigrant communities (Pelle and Pappadà, 2021; Vacca, 2020)—baseline density and modularity parameters may shift substantially. Future research should apply our empirical decision tree across diverse non-collegiate populations to evaluate the institutional portability and sensitivity of these topological cutoffs.

Second, our survey instrument relied on open-ended name generators that elicited up to 20 to 25 active alters per wave. While combining eight survey waves over three years generated expansive cumulative networks ( $N = 701$ ; 35,912 nominations; 174,748 alter–alter evaluations), bounded name-generator designs inevitably focus respondents on salient, active relational ties, potentially truncating the distant periphery of weak acquaintances (Granovetter, 1973). Future investigations can address this constraint by pairing survey-based name generators with passive digital trace data—such as smartphone communication metadata, Bluetooth proximity sensors, or institutional co-registration graphs (Liu et al., 2018)—to test whether including peripheral contacts alters community partitioning and boundary thresholds.

Third, although our longitudinal Markov models capture directional semester-to-semester state transitions across 1,900 intervals, the cross-sectional evaluation of functional social support within network types remains observational. Consequently, parameter estimates represent conditional associations rather than strictly identified causal effects. Does embedding within a tightly knit Regular Dense enclave actively cultivate high-trust multiplexity through normative closure and mutual surveillance (Coleman, 1988), or do individuals with high psychological needs for emotional intimacy selectively prune non-multiplex alters until a cohesive clique remains? Future studies can disentangle these dynamic selection and influence

mechanisms by deploying continuous-time stochastic actor-oriented models (SAOMs) or cross-lagged panel models tracking support exchange alongside topological evolution over time.

Finally, while this study establishes the topological taxonomy, classification rules, developmental transitions, and functional support profiles of personal networks, it does not trace downstream life-course outcomes. Future research should investigate how sustained occupancy in specific network regimes shapes subsequent academic performance, psychological well-being, feelings of loneliness, and early occupational trajectories upon transition into the post-collegiate labor market.

### 5.3 Theoretical Implications: Toward a Structural Sociology of Personal Networks

The empirical findings presented here carry significant theoretical implications for the structural analysis of personal networks and contemporary social capital theory.

Most centrally, our results resolve the longstanding tension between deductive theoretical taxonomies and inductive algorithmic clustering (Bidart et al., 2018; González-Casado et al., 2024; Vacca, 2020). For decades, researchers remained caught between transparent but arbitrary heuristic cutoffs derived from small, idiosyncratic samples (Bidart et al., 2018) and unsupervised clustering algorithms that partition networks without yielding portable, interpretable rules (Vacca, 2020). By uniting unsupervised  $k$ -means clustering with recursive partitioning classification trees, our analytical workflow demonstrates that egocentric relational space can be mapped with mathematical rigor while producing explicit, transparent decision boundaries that other scholars can operationalize directly in comparative datasets.

Furthermore, our findings illuminate the dynamic, temporal nature of personal social capital. Classical network theory often treats personal networks as static structures or enduring individual attributes (Fischer, 1982; McCarty, 2002). Our longitudinal transition models challenge this static view, demonstrating that personal networks are developmental systems that oscillate between localized cohesive enclaves and expansive modular bridging pathways. Students do not inhabit permanent structural categories; rather, they transition across network regimes as institutional demands, residential reassignments, and developmental milestones reshape their social ecologies.

Finally, our findings reconcile the structural and compositional traditions in egocentric network analysis. Rather than viewing graph topology and relational content as competing paradigms (Antonucci et al., 2013; McCarty, 2002; Offer and Fischer, 2018), our results show that they represent two sides of the same relational coin. Egocentric network geometry



establishes the physical and communicative constraints that govern the bandwidth of social support: expansive, modular topologies maximize bridging social capital and access to novel information (Burt, 1992; Granovetter, 1973), but do so at the cost of relational multiplexity and emotional depth. Conversely, small, cohesive enclaves sacrifice structural reach in order to cultivate dense mutual trust, emotional solidarity, and multi-functional safety nets (Coleman, 1988; Lin, 2001). By integrating topological geometry, empirical decision rules, longitudinal transitions, and functional support provision, this study provides a unified structural foundation for understanding how personal networks organize interpersonal life.

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